

A Feature-Based Corpus Typology of Perfect-Like Grams

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1 Introduction & Motivation

1.1 Project Overview

Perfect-like grams (PLGs) show systematic cross-linguistic patterns shaped by morphology-semantics mapping, language-specific TAM systems, and potentially other interacting factors. This project seeks to develop a feature-based typological model using parallel corpora and computational pipelines to explore cross-linguistic correlations among these factors. To test this framework, I use machine learning classifiers to predict PLGs across languages. Their outputs are then analyzed with inferential statistics, weighted error diagnostics, and feature-based semantic mapping, producing results that are both computationally rigorous and linguistically interpretable. While the methodology has a computational backbone (data science, NLP, machine learning), the ultimate goal is linguistic: refining our understanding of PLGs, highlighting underdescribed variation, and connecting computational tools back to linguistic theory.

This dissertation is comprised of three interlocking projects. Together they (i) evaluate the ability of an ontological feature schema to capture cross-linguistic distributions and interpretations of PLGs, including comparison with vector-based distributional models such as BERT and S-BERT; (ii) uncover novel insights into the forms and meanings of PLGs and into typological comparison more broadly; and (iii) test the scalability of this approach to generalize to new languages and unseen data.

1.2 Aims & Research Questions

The project pursues four mutually reinforcing aims:

1. **Linguistic insight.** The core aim of the thesis is to leverage computational methodologies to better understand and explain linguistic phenomena. The object of study, cross-linguistic perfect-like grams (PLGs), is a rich domain for this undertaking as it sits at the junction of the theory and typology tense-aspect-mood (TAM) research. This project focuses on using classification experiments alongside semantic mapping tools to test how well a theory- and typology-informed ontological feature schema can capture cross-linguistic patterns in PLGs through observing both the predictive capabilities of such a linguistically-grounded schema as well as clustering patterns of meanings, morphological expression, language-specific TAM system, and other factors that may be at play. This will be accomplished by ensuring that the computational framework remains linguistically informed, while also generating new evidence to inform debates in TAM typology, language description, and theory.
2. **Feature schema development.** Central to the project is the gram and iterative refinement of a cross-linguistic PLG feature inventory, grounded in TAM literature and corpus evidence,

with diagnostics for interpreting feature values. Corpora such as the Bible and TED Talks will provide initial data for testing and refinement, and scalability testing will incorporate data from the DoReCo corpora.

3. **Cross-disciplinary integration.** This project further seeks to demonstrate how computational pipelines can both sharpen and address linguistic questions while linguistic theory can enrich computational models, producing a roadmap for ongoing, cross-disciplinary collaboration between linguistics and NLP.
4. **Ontological innovation.** Ultimately, the goal of this research program is to develop a prototype for applying ontology-driven schemas to other domains (e.g. clause type contrasts, TAM ambiguity, low-resource languages). This involves piloting an ontological feature schema in machine learning, encoding semantic featural distinctions such as experiential versus resultative readings into a structured, machine-readable ontology. This allows model performance to be interpreted in typologically and semantically meaningful terms, establishing a pathway to future scalability and boarder applications.

These aims coalesce into two central questions, which guide the project as a whole.

- To what extent can a feature-based semantic ontology, developed on the basis of aligned corpus data, but annotated on a single language, model cross-linguistic variation in PLGs?
- Does the application of such a schema reveal typologically consistent contrasts (e.g. experientiality, evidentiality, so-called “current relevance”) across languages that differ morphosyntactically and genealogically, providing a scalable framework for typological comparison, semantic annotation, and temporally complex meanings?

Concomitant operational questions include:

- Is the schema internally coherent and usable across languages?
- Can the features be clearly defined and consistently annotated?
- Do these features effectively capture the typological diversity of PLGs across languages?
- How are these features instantiated (morphologically/morphosyntactically) across languages, what systematic profiles emerge, and what correlations can be drawn between these factors?
- Are there recurring patterns or shared strategies across unrelated languages?

Together, these questions position the project at the intersection of typology, description, and computation, asking not only whether PLGs form a coherent cross-linguistic class (cf. Dahl & Wälchli 2016), but also whether a feature-based approach can make that coherence (or lack thereof) explicit, interpretable, and empirically testable.

1.3 Background & Literature Review

1.3.1 Perfect-Like grams: A Cross-Linguistic Puzzle

Perfect-like grams (PLGs) sit at the intersection of tense, aspect, modality (TAM), and information structure, yet resist tidy definition even within individual languages, let alone across them. While many languages possess TAM categories that have been compared to the English

present perfect (e.g. French *passé composé*, German *Perfekt*, Turkish evidentials, Mandarin *guò*), the semantic contributions of these forms vary widely, often defying one-to-one translation and challenging conventional comparative frameworks (Bertrand et al., 2022; Dahl, 2000; Dahl & Wälchli, 2016).

Despite their apparent shared characteristics, such as links to past events, non-narrative temporal contours, or present relevance (“current relevance”, “discourse salience”, etc.), PLGs have long been recognized as resistant to neat categorizations (Plungian, 2011; Schaden, 2009), a quality reflected in the ongoing theoretical debate over their core semantics. For example, even closely related European languages diverge sharply in how PLGs interact with tense and aspect systems. As shown in (1a, b), in French and German, PLGs are compatible with definite temporal adverbial modification (“DTAM”), while in English and Spanish (examples (1c, d)) they are not¹.

- (1) a. English
I have read this book (*yesterday).
- b. Spanish
He leído este libro (*ayer)
have.1SG.PRES read.PTCP this book (*yesterday)
‘I have read this book (*yesterday).’
- c. German
Ich habe diese Buch gestern gelesen
I have.1SG.PRES this book yesterday read.PTCP
‘I [have] read this book yesterday.’
- d. French
J’ai lu ce livre hier
I.have1SG.PRES read.PTCP this book yesterday
‘I [have] read this book yesterday.’

Such contrasts show how easily even typologically close languages break apart under fine-grained semantic diagnostics, highlighting the limits of categorical comparison.

Even within a single language, multiple PLG strategies can coexist, some overlapping with English perfects and others diverging in subtle but systematic ways. In Japanese for instance, as shown in (2), [verb + *koto ga aru*] expresses an experiential reading similar to English perfects, while the simple past *-ta* form can yield resultative interpretations in certain contexts, as shown in (2b) below². Georgian likewise illustrates this point: the Synthetic Perfect carries evidential force, while the Periphrastic Perfect does not, such that (3a) is only felicitous in contexts where the speaker wishes to convey a particular stance with respect to the source of knowledge (in the Georgian Synthetic Perfect, visual evidence or second hand – including traditional – knowledge), where as (3b) is evidentially neutral, and must explicit include an adverbial (such as *turme* ‘apparently’ or *rogorts chans* ‘it seems that’) in order to convey a similar meaning³.

¹This contrast is widely discussed in the literature as the “Present Perfect Puzzle” (e.g. Klein 1992; Iatridou, Anagnostopoulou, & Izvorski 2001; Pancheva 2004; many others).

²Sakai 1981; Ogihara 1989

³Makharoblidze (2012); Hewitt (1996); many others

- (2) a. Experiential *koto ga aru*
 nihon ni itta koto ga aru
 Japan to go.PAST NMLZ PTCL be
 ‘I have been to Japan.’ (Lit. ‘there is an event in which I went to Japan’)
- b. Resultative *-ta*
 tsukareta
 tire.PAST
 ‘I am tired.’ (Lit. ‘I [became] tired’, ‘I [have become] tired’)
- (3) Georgian Synthetic and Periphrastic Perfects
- a. Synthetic Perfect with grammatical evidentiality
 Anzor-s es ts’igni ts’a-u-k’itkh-av-s
 Anzor-DAT this book PV-VS-read-TS-3SG
 ‘Anzor has apparently read this book.’
- b. Periphrastic perfect with neutral evidentiality
 Turme / rogorts chans Anzor-s es ts’igni ts’a-k’itkh-ul-i akvs
 apparently / it seems that Anzor-DAT this book PV-read-PTCP-NOM have.3SG.PRES
 ‘Apparently/it seems that Anzor has read this book.’

The DTAM contrast illustrates how PLGs diverge across closely related languages, while the Japanese and Georgian examples show how multiple PLG strategies can coexist within a single language. Together, these patterns highlight the need for methods that go beyond categorical comparison and instead track underlying features. By moving from categorical comparisons to feature-based analysis across large, multilingual corpora, this project aims to reveal cross-linguistic contrasts and commonalities that would be invisible in small-scale elicitation or in one-to-one translation equivalence. This typological diversity, both across and within languages, is precisely what makes PLGs a compelling object of typological and computational inquiry: their semantic subtlety and morphosyntactic variability pose significant challenges for formal theory, while offering a valuable testing ground for empirically grounded, feature-driven models.

To situate these puzzles within broader debates, we turn to the core readings canonically attributed to the perfect: experiential (emphasizing that an event has occurred at least once before the speech time), resultative (focusing on a present state that results from a past event), universal (a state or event beginning prior to speech time and continuing up to and/or through it), and recent past (a situation occurring shortly before speech time). Each of these readings is exemplified in (4)

- (4) a. **Experiential:** She has read this book (before / three times / at some point).
 b. **Resultative:** The sun has set (and now it is dark).
 c. **Universal:** We have lived here (for three years / since 2007).
 d. **Recent Past:** The letter has just arrived.

1.3.2 Theoretical Foundations: Core Semantic Readings of the Perfect

As stated above, PLGs have typically been associated with a small set of recurring interpretations. These readings, while not always discrete in practice, have provided a useful descriptive

framework for comparing TAM grams across languages (Comrie, 1976; Klein, 1992; McCoard, 1976; Mittwoch, 2008). As seen in studies such as Dahl (1985) and Plungian & Auwera (2006), these categories can be used to distinguish PLGs across genealogically diverse languages.

In the context of cross-linguistic PLG comparison, experientiality offers a testable feature, namely, the presence of quantified event recurrence such as in *I have eaten durin [before, three times, etc.]*, that can guide corpus annotation and comparison. Resultatives, by contrast, imply a change-of-state event whose resultant state holds at reference time (e.g. the window has broken, the butter has melted). Universals extend the predicate across a continuous time span up to the reference point (e.g. I have lived here for five years, she has always had brown hair). While these categories reflect theoretical abstraction, they remain widely used in typological and computational work, e.g. perfect-aspect tag distinctions in the Universal Dependencies project or in multilingual NLP tasks like temporal relation extraction (Zhao 2022a).

Departing from traditional typological treatments, the current project takes these readings not as mutually exclusive categories, but rather as semantic features that potentially co-occur. For instance, a single gram may simultaneously express experiential and resultative features, a pattern that resists categorization in existing discrete typologies but aligns with a feature-based representation. This approach reflects a move away from discrete categories toward feature-based modeling approaches (Bertrand et al., 2022; Haspelmath, 2010a,b).

1.3.3 Cross-Linguistic Approaches: From Nebulous Categories to Predictive Features

The typological complexity of PLGs has been extensively documented in survey work and large-scale comparative projects. Seminal work by Dahl (2000) and Dahl & Velupillai (2013) established broad patterns in TAM systems across languages, while more recent studies (Bertrand et al., 2022; Bowler & Ozkan, 2017; Zhao, 2022a,b) have emphasized the gradient, overlapping nature of the semantic denotations of PLG-like grams (e.g. perfect-like grams in Turkish and Bulgarian encode both anteriority and evidentiality, but differ in source and speaker orientation⁴) across genetically and areally diverse systems.

A key insight from this literature is that cross-linguistic variation is often more a matter of degree than kind, which is to say that many languages encode 'perfect-like' meanings, such as recurrence, relevance to reference time, or resultant state, using tense-aspect combinations, adverbials, or particles, rather than grammaticalizing a distinct perfect gram. The overlap or convergence between perfects and iamitives (Dahl & Velupillai 2013), as well as the often fuzzy boundary between resultative and evidential readings (Plungian 2011), reinforce the need for modeling strategies that represent cross-linguistic polysemy without reducing diverse grams to a single cross-linguistic category.

This has led to a recent wave of work attempting to identify and formalize cross-linguistically predictive features of TAM categories (e.g. iterative experientiality, persistent state, or temporal anchoring) from corpus data and annotated language resources, a trend this project builds upon to enable more precise and nuanced cross-linguistic comparison and linguistic data annotation. Of particular relevance is the work of van der Klis et al. (2017, 2021), LeBruyn et al. (2019), and Zhao (2022, 2022), who propose annotation schemas and classification models that incorporate semantic, pragmatic, and discourse-related factors. However, these models often rely on fixed categorical inventories of TAM labels (e.g. Resultative, Perfective), rather than decomposing grams into cross-linguistic semantic features and studying how those features cluster. This project diverges by embracing a scalable, feature-oriented ontology informed by cross-linguistic corpus data, which

⁴Bowler & Ozkan (2017)

organizes TAM grams through cross-linguistically predictive semantic features and their clustering patterns, rather than through fixed categorical labels.

1.3.4 Methodological Orientation: Toward Semantics-Lite Ontologies

The goal of this project is to develop a usable, interpretable feature set that can track variation in PLGs across typologically distinct systems, and that can be productively deployed in both research and applied NLP contexts, facilitating dialogue between these domains. While indebted to formal semantic accounts of the perfect (Portner 2003, 2011; Kratzer 2004; Iatridou, Anagnostopoulou, & Izvorski 2001), the project is not primarily concerned with extending that tradition, but rather with building a practical feature ontology for cross-linguistic comparison and annotation.

In doing so, the project draws a working distinction between inherent semantic features, i.e. those conventionally encoded in the truth conditional semantics of a gram⁵, and context-dependent interpretations, which may arise from pragmatic inference or usage context. This distinction, while not exhaustively formalized in the pilot, informs both feature design and annotation logic, and reflects ongoing conversations in formal semantics around the boundary between encoded and inferred meaning.

This approach aligns with recent calls in linguistic typology and computational semantics for hybrid models, ones that acknowledge both the utility of comparative concepts and the importance of language-specific, gram-based patterns (Haspelmath, 2010a,b, 2016). By grounding feature definitions in observable diagnostics (e.g. compatibility with temporal adverbs, stative versus eventive predicates, etc.), and by testing them against real corpus data, this work aims to strike a balance between cross-linguistic abstraction and alignment with attested usage patterns across languages. In doing so, it hopes to extend the toolkit available for modeling temporally complex meaning in natural language, while also providing a more interpretable alternative to opaque, large-model-derived representations (cf. Chen et al. 2021).

2 Pilot Experiment

This section presents the completed pilot experiment which informs and shapes the following work of the project.

2.1 Motivation & Goals

Perfect-like grams (PLGs) pose challenges for both linguistic typology and computational modeling. Their semantic and morphosyntactic complexity (such as the interaction of anteriority with evidentiality, or variation in auxiliary selection across contexts) makes them difficult to capture through traditional cross-linguistic comparison, while their variability (such as context-sensitive shifts between experiential and resultative readings) also poses difficulties for current NLP approaches, which often rely on surface-level cues or discrete category labels in tasks like tense detection, temporal relation classification, and cross-lingual parsing. These systems may misinterpret perfect-like forms as simple pasts, or fail to distinguish between readings that differ subtly in temporal or evidential nuance.

The three principle aims of the pilot were to assess whether (i) the proposed schema, a feature-based annotation framework for PLGs (see §2.2.1) could be consistently applied across languages;

⁵E.g. a gram may inherently encode anteriority, but be interpreted as evidential or experiential depending on discourse context.